```
!pip install transformers
!pip install datasets
!pip install scikit-learn
```

```
: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.0)
: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.5.0)
: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (202
: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers)
: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28
: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.1)
: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformer
: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers
: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transforme
: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->tran
: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers
: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=
: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (8
: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transforme
: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1
: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transfor
: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=
: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0
: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingfac
: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.25.0)
: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->da
: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.5.0)
: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->dat
: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (1.3.2)
: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (0.28.1)
: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (0.24.1)
: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (4.15
: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.4)
: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11)
: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.5.0)
: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2026.2.25)
: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<
: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3
: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0
```

```
import torch
import numpy as np
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("imdb")

print(dataset)
```

```
·/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
 secret `HF_TOKEN` does not exist in your Colab secrets.
authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/toker
 will be able to reuse this secret in all of your notebooks.
ase note that authentication is recommended but still optional to access public models or datasets.
arnings.warn(
DME.md:       7.81k/?  [00:00<00:00, 386kB/s]
ning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits a
NING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_
_text/train-00000-of-00001.parquet: 100%  21.0M/21.0M  [00:01<00:00, 105MB/s]
_text/test-00000-of-00001.parquet: 100%  20.5M/20.5M  [00:00<00:00, 102MB/s]
_text/unsupervised-00000-of-00001.p(…): 100%  42.0M/42.0M  [00:01<00:00, 68.4MB/s]
erating train split: 100%  25000/25000  [00:00<00:00, 68852.93 examples/s]
erating test split: 100%  25000/25000  [00:00<00:00, 104136.70 examples/s]
erating unsupervised split: 100%  50000/50000  [00:00<00:00, 102092.45 examples/s]
asetDict({
 train: Dataset({
     features: ['text', 'label'],
     num_rows: 25000
 })
 test: Dataset({
     features: ['text', 'label'],
     num_rows: 25000
 })
 unsupervised: Dataset({
     features: ['text', 'label'],
     num_rows: 50000
 })
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")

def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

tokenized_dataset = dataset.map(tokenize_function, batched=True)
```

```
config.json: 100%  483/483  [00:00<00:00, 10.4kB/s]
tokenizer_config.json: 100%  48.0/48.0  [00:00<00:00, 3.91kB/s]
vocab.txt:       232k/?  [00:00<00:00, 4.97MB/s]
tokenizer.json:       466k/?  [00:00<00:00, 4.87MB/s]
Map: 100%  25000/25000  [00:49<00:00, 768.84 examples/s]
Map: 100%  25000/25000  [00:35<00:00, 583.52 examples/s]
Map: 100%  50000/50000  [01:13<00:00, 825.73 examples/s]
```

```
small_train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(2000))
small_test_dataset = tokenized_dataset["test"].shuffle(seed=42).select(range(1000))
```

```
small_train_dataset = small_train_dataset.rename_column("label", "labels")
small_test_dataset = small_test_dataset.rename_column("label", "labels")

small_train_dataset.set_format("torch", columns=["input_ids", "attention_mask", "labels"])
small_test_dataset.set_format("torch", columns=["input_ids", "attention_mask", "labels"])
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```
model.safetensors: 100%                                    268M/268M [00:06<00:00, 70.9MB/s]

Loading weights: 100%                 100/100 [00:00<00:00, 277.17it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_projector.bias     | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
classifier.weight        | MISSING    |
pre_classifier.bias      | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your down
```

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    eval_strategy="epoch",
    logging_dir="./logs"
)
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=small_train_dataset,
    eval_dataset=small_test_dataset,
    compute_metrics=compute_metrics
)
```